

Transformers, power flow analysis part 2

ELEC0447 - Analysis of electric power and energy systems

Bertrand Cornélusse

September 7, 2026

Overview I

1. The Transformer

2. Transformers in the power flow analysis

What will we learn today?

- ▶ The power transformer
- ▶ The next part of power flow analysis: how to include transformers and transformers with tap changers

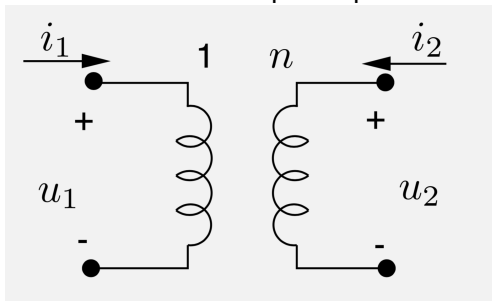
You will be able to do exercises 6.2, 6.3, 6.4 from Ned Mohan's book.

The Transformer

The background of the slide features a minimalist design with teal-colored geometric shapes. Two large teal triangles point upwards from the bottom, meeting at a central point. Below this meeting point, there is a smaller, darker teal triangle pointing downwards. The remaining space at the top of the slide is white.

Basic model of a transformer

A (single phase) transformer is made of two magnetically coupled **coils** or **windings**. An ideal transformer is a two-port represented as



with

$$u_2 = nu_1$$
$$i_2 = -\frac{1}{n}i_1$$

Usages

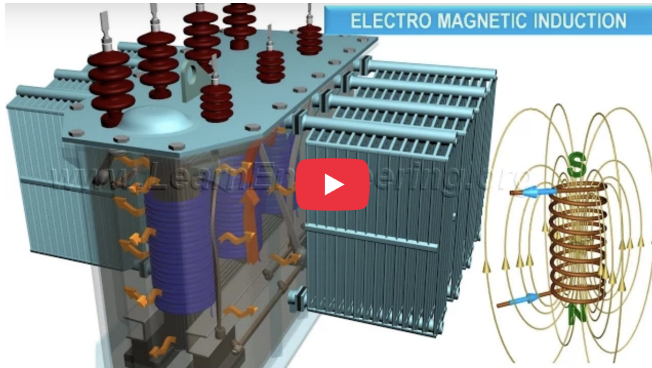
In power systems, transformers are mainly used to transmit power over long distances by changing the voltage level, thus decreasing the current for a given power level. The voltage level of a synchronous generator is around 20kV.

Voltage is changed around five times between generation and load.

It is also used to

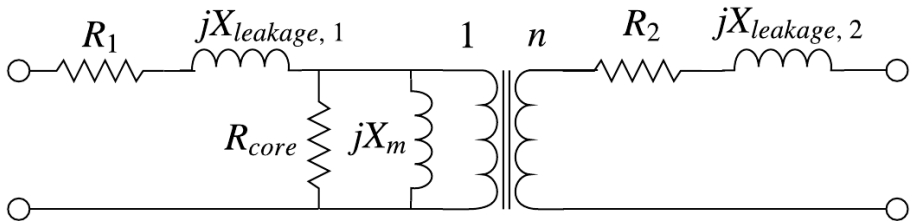
- ▶ **measure** currents and voltages,
- ▶ electrically **isolate** parts of a circuit (not the auto-transformer we will see),
- ▶ and **match** impedances.

How does a transformer work?



[Link to the video](#)

Non-ideal model I



The ideal model is complemented by elements

Non-ideal model II

- ▶ X_m that models the magnetizing inductance
- ▶ $X_{leakage,i}$ that models the flux not captured by the core on side i
- ▶ R_{core} that models eddy current and hysteresis losses, i.e., losses in the iron core
- ▶ R_1 and R_2 that model (coil) copper losses

Parameters are either given in the datasheet or obtained by open-circuit and short-circuit tests.

The core is usually **laminated** to decrease losses.

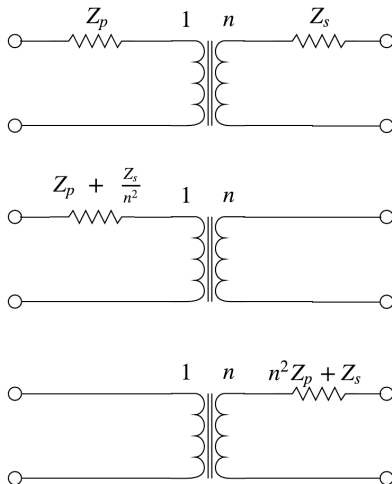
Simplification

The excitation current, the sum of the currents in R_{core} and X_m , is often neglected, leading to a simpler non-ideal model, and the series impedances can be transferred from one side to the other, with

$$Z_p = R_1 + jX_{leakage,1}$$

and

$$Z_s = R_2 + jX_{leakage,2}$$



Per unit representation I

Let's consider the rated voltages and currents on both sides of the (ideal) transformer as base values. As

$$V_{s,base} = nV_{p,base}$$

and

$$I_{p,base} = nI_{s,base},$$

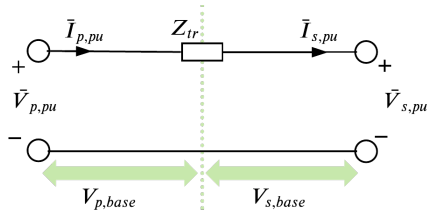
the **MVA base is the same on both sides**, and thus

$$Z_{s,base} = n^2 Z_{p,base}$$

Per unit representation II

Hence, **in per unit, the transformer can be replaced by a single impedance**

$$Z_{tr} = \frac{Z_p}{Z_{p,base}} + \frac{Z_s}{Z_{s,base}}.$$



Per unit representation III

Thus we have also that

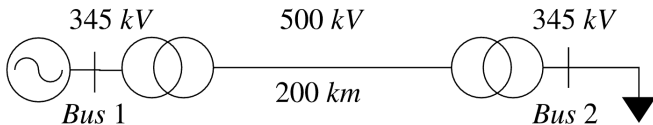
$$\begin{aligned} Z_{tr} &= \frac{Z_p + Z_s/n^2}{Z_{p,base}} \\ &= \frac{n^2 Z_p + Z_s}{Z_{s,base}} \end{aligned}$$

i.e. **the impedance is the same whether we see it from the primary or the secondary side, although the voltage bases differ.**

Also, if the three-phase transformer is **wye-delta** connected, a **30° phase shift** must be applied (more on this later).

Illustration of Per-Unit normalization I

This is Example 6.1 from the reference book. Consider the one-line diagram



with

- ▶ a 200 km line with $R = 0.029\Omega/km$, $X = 0.326\Omega/km$, neglected shunt impedances
- ▶ two transformers with a leakage reactance of $0.2pu$ in the (500 kV, 1000 MVA) base, and losses neglected.

Illustration of Per-Unit normalization II

What is the equivalent model in a (345 kV, 100 MVA) base?

In the (500 kV, 1000 MVA) base:

- ▶ $Z_{line,pu} = 200 \times (0.029 + j0.326)/(500^2/1000) = 0.0232 + j0.2608pu$
- ▶ hence, the total impedance between buses 1 and 2 is

$$Z_{12} = 0.0232 + j0.2608 + 2 * j0.2pu = 0.0232 + j0.6608pu$$

In the (345 kV, 100 MVA) base

- ▶ the pu value of the impedance is the same in the (500 kV, 1000 MVA) and (345 kV, 1000 MVA) bases!

Illustration of Per-Unit normalization III

- ▶ Why? since we can transfer the impedance from one side of each transformer to the other, cf. a previous remark.
- ▶ if we now change the MVA base to 100 MVA,

$$Z_{12} = (0.0232 + j0.6608) \times (100/1000)pu = 0.00232 + j0.06608pu$$

since the base impedance is proportional to the inverse of the MVA base.

Efficiency

The efficiency expressed in % is

$$100 \times \frac{P_{output}}{P_{input}} = 100 \times \left(1 - \frac{P_{losses}}{P_{input}} \right)$$

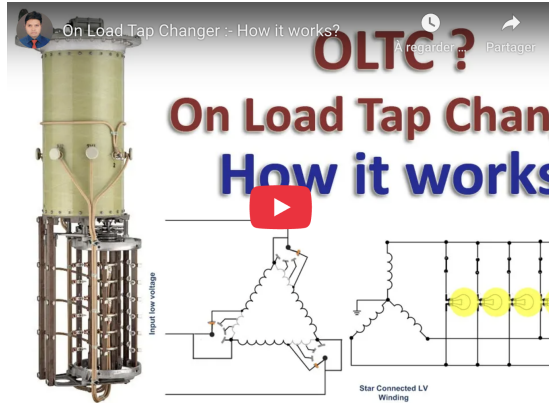
Remarks:

- ▶ maximal when loaded such that copper losses = iron losses (cancel derivative of efficiency w.r.t current)
- ▶ Around **99.5 %** in large power transformers at full load.

Tap changers

- ▶ Some transformers are equipped with a system allowing the $1 : n$ ratio to be changed
- ▶ The ability to change the tap under load is called load tap changer (LTC) or on-load tap changer (OLTC)
- ▶ This is mainly used for voltage control
- ▶ It is usually implemented using auto-transformers
- ▶ We will see later on how to include this in the power flow analysis

Tap changing transformers



[Link to the video](#)

Auto-transformers

The two windings (of the same phase) are connected in series, without galvanic insulation.
They are commonly used when the ratio is limited.

Advantages:

- ▶ Physically smaller
- ▶ less costly (less copper)
- ▶ higher efficiency
- ▶ easy to implement tap changes
- ▶ "solid" earth grounding

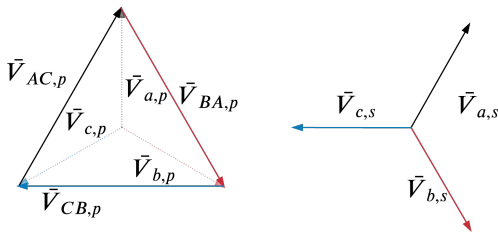
Disadvantages:

- ▶ no electrical insulation
- ▶ higher short circuit current
- ▶ full voltage at secondary if it breaks (in case of a step down)

Phase shift in delta-star transformers

The star part has n times the number of turns of the delta part (primary side).
Let's reason on phase a ,

- ▶ Voltage $\bar{V}_{a,s}$ is on the same core as $\bar{V}_{AC,p} = \sqrt{3}\bar{V}_{a,p}\angle -30^\circ$ where $\bar{V}_{a,p}$ is the (virtual) phase-neutral voltage on the primary side.
- ▶ Since $\bar{V}_{a,s} = n\bar{V}_{AC,p}$,
 $\bar{V}_{a,s} = n\sqrt{3}\bar{V}_{a,p}\angle -30^\circ$

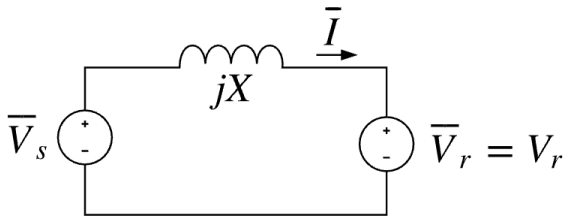


We **gain a $\sqrt{3}$ factor** in the amplification, and a **lagging phase shift of 30°** .

The same reasoning holds for phases b and c .

Power flow regulation by phase shifting I

We have seen that active power flows are dictated by the voltage magnitudes but also the sine of the angle difference between buses:



$$\begin{aligned} S_r &= \bar{V}_r \bar{I}^* = V_r \left(\frac{V_s \angle -\delta - V_r}{-jX} \right) \\ &= \frac{V_s V_r \sin \delta}{X} + j \frac{V_s V_r \cos \delta - V_r^2}{X} \end{aligned}$$

δ is the angle between $\bar{V}_r$ and $\bar{V}_s$

Power flow regulation by phase shifting II

If we have a device that can generate an adjustable phase shift, we can control the power flows. This is the purpose of **phase-shifting transformers**.

In practice phase shifting is achieved by "combining the signal with a fraction of itself shifted by 90° ". For the details of how it is implemented or modeled, see

- ▶ Wikipedia
- ▶ Section 5.7 of Weedy or ELEC0014.
- ▶ ENTSO-E - Phase Shift Transformers Modelling, Version 1.0.0, May 2015

Example: phase shifting transformers on the borders of Belgium I



380/380 kV, in series with:

1. line Zandvliet (B) - Borssele (NL)
and Zandvliet (B) - Geertruidenberg (NL)
2. line Meerhout (B) - Maasbracht (NL)
3. line Gramme (B) - Maasbracht (NL)
 - ▶ nominal power
 $3V_N I_{\max} = 1400 \text{ MVA}$
 - ▶ phase shift adjustment: 35
positions, $+17/ - 17 \times 1.5^\circ$ (at
no load)

Example: phase shifting transformers on the borders of Belgium II

220/150 kV :

- ▶ in series with the Chooz (F) - Monceau (B) line nominal power: 400 MVA
- ▶ in-phase adjustment: 21 positions, $+10/ - 10 \times 1.5\%$
- ▶ quadrature adjustment: 21 positions, $+10/ - 10 \times 1.2^\circ$

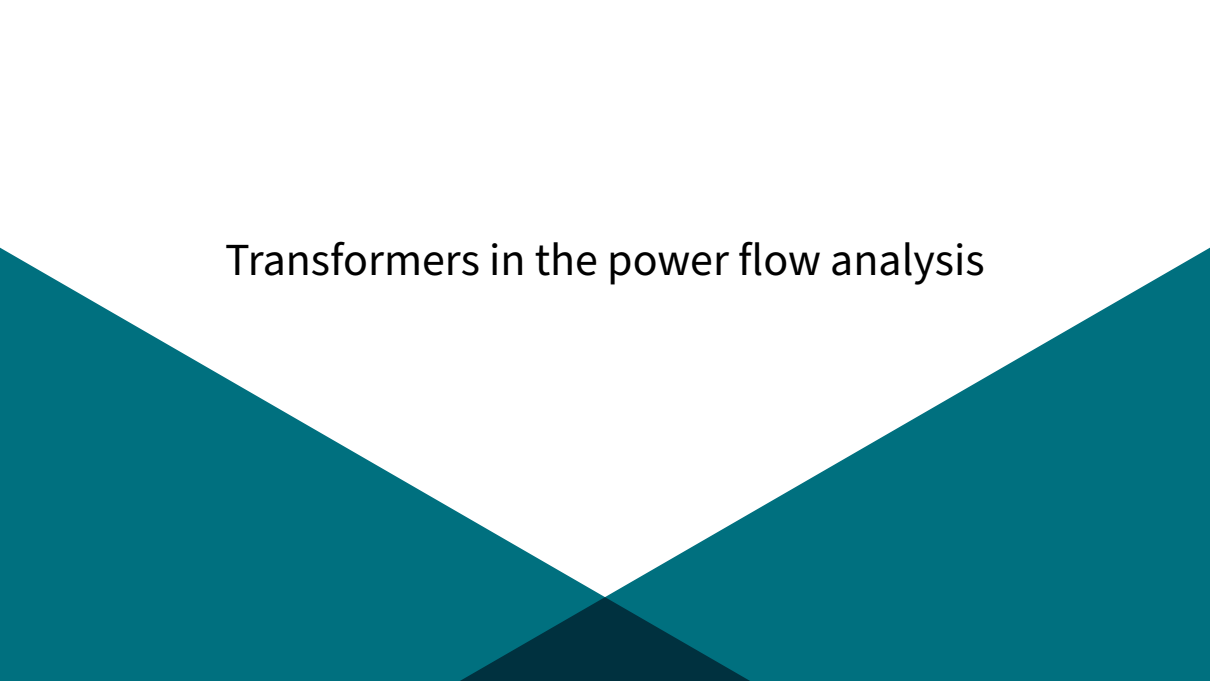
Remarks

In **three-phase operation**,

- ▶ either there are three separate single-phase transformers (easier to fix when there is a problem on a phase, more modular)
- ▶ or a **three-phase transformer**, that is a single core with three auto-transformers on it, cf. the video at the beginning of this presentation (cheaper, lighter core and less copper).

Some transformers called **three-winding transformers** are equipped with a third winding (not to be confused with a three-phase transformer) that is used for auxiliary purposes (feeding auxiliary devices e.g., fans, providing reactive power support, ...).

Transformers in the power flow analysis

The background of the slide features a white central area where the text is located. This area is framed by large, teal-colored geometric shapes. On the left, a teal triangle points downwards towards the center. On the right, another teal triangle points upwards towards the center. These two triangles meet at a point at the bottom center, where a darker teal triangle points upwards, creating a symmetrical, abstract design.

Transformer without regulation

A transformer, in the per-unit representation, can thus be represented

- ▶ as a two-port if the shunt admittance is considered
- ▶ as a simple series leakage impedance if the shunt admittance is neglected

It is a bit harder to model controllable taps for voltage and flow control.

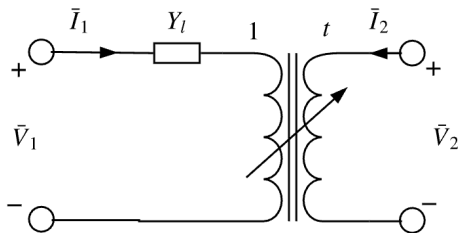
Representing taps and phase shifts I

Let Y_l be the leakage admittance and t be the off-nominal turns ratio:

- ▶ if $0 < t \leq 1$, this corresponds to a simple tap-changer
- ▶ if $0 < |t| \leq 1$ but is complex, then this is a phase-shifter ($\angle t < \pi/2$)

We have

$$\bar{I}_1 = \left(\bar{V}_1 - \frac{\bar{V}_2}{t} \right) Y_l$$



Representing taps and phase shifts II

and since $\frac{\bar{V}_2}{t} \bar{I}_1^* = -\bar{V}_2 \bar{I}_2^*$ by energy conservation

$$\bar{I}_2 = -\frac{\bar{I}_1}{t^*} = -\bar{V}_1 \frac{Y_l}{t^*} + \bar{V}_2 \frac{Y_l}{|t|^2}$$

Thus tap and phase shift can be represented by the admittance matrix

$$\begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} Y_l & -\frac{Y_l}{t} \\ -\frac{Y_l}{t^*} & \frac{Y_l}{|t|^2} \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix}$$

Representing taps and phase shifts III

- ▶ if $0 < t \leq 1$, this can be represented as a π two-port
- ▶ if $0 < |t| \leq 1$ but is complex, this is not the case

In the power flow analysis you must **pay attention to this when constructing the system-wide admittance matrix**. As an exercise, let's add a phase-shifting transformer to our 3-bus example.

- ▶ [Link to the example in PandaPower](#)
- ▶ [Link to the Newton-Raphson example](#)